flashjet

GPU-native jet clustering with the full merge history

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ML4Jets 2026 · Vienna

September 2026

Benchmarks: **ATLAS Open Data** (CC0). Closures: analytic predictions on toy showers.

Outline

Jet clustering, and flashjet

Reconstruction is moving to accelerators — clustering has not

Reconstruction is steadily becoming **heterogeneous**: as much of the event as possible is expected to run on the accelerator.

CMS has run GPUs in the High-Level Trigger since the start of Run 3 — **∼30 % of HLT reconstruction** is offloaded (pixel local reconstruction, pixel-only tracks and vertices, calorimeter local reconstruction), for a **∼25 % reduction** in total HLT time.

Tracking, vertexing and calorimetry have all been ported. **Jet clustering has not** — it is one of the few steps still bound to the CPU. [CMS HLT, Run 3]

Why it is still on the CPU

Sequential recombination is **IRC safe** — which is why it survived — but each merge **depends on the one before it**. That data dependence is exactly what does not map onto a GPU in the obvious way.

Where it hurts today

- reclustering jets **inside a training loop**
- scanning the **jet radius**
- accessing **substructure** dynamically

Each pays a **host** $\leftrightarrow$ **device transfer**, over and over.

flashjet

An **open-source** library bringing k_t, Cambridge–Aachen and anti-k_t natively to the GPU through custom **Triton** kernels.

Why Triton. It is the GPU language of the PyTorch ecosystem, so the kernel **compiles and autotunes for the hardware at hand** and drops into tensor pipelines directly — with **no separate CUDA build** and no hand-written device code to maintain.

Execution model. One kernel program clusters **one event entirely on chip**; the whole batch goes out in a **single fused launch**. Nothing returns to the host.

Jets, clustering histories and derived substructure all come back as **tensors**.

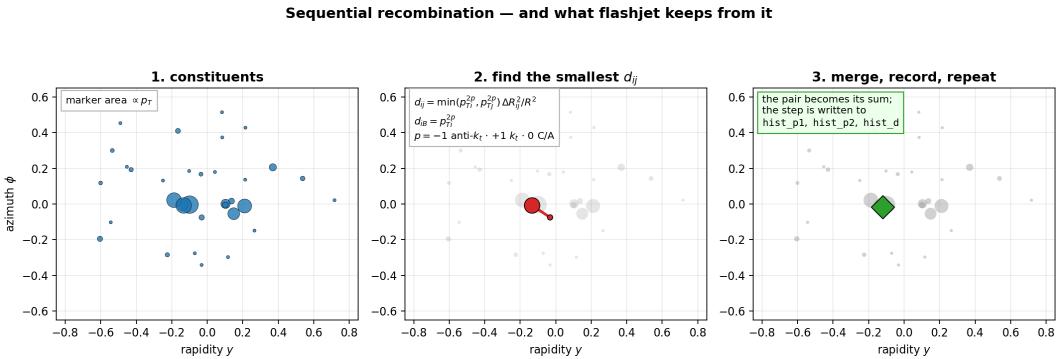
What makes it tractable

The Cacciari–Salam nearest-neighbour lemma: the global d_{ij} minimum is always realised at some particle's **geometric** nearest neighbour. Each slot carries its own NN, so a merge invalidates only $O(1)$ rows on average — $O(N^2)$ **per event instead of** $O(N^3)$.

Engineering

- bandwidth-bound: the NN update and the **next argmin** are fused into one sweep
- **branchless** pair-vs-beam, via masked stores
- launch config autotuned **once per GPU model** and cached, so runs stay **reproducible**

How sequential recombination works



Repeat until nothing is left: if the smallest distance is a d_{ij}, merge that pair; if it is a d_{iB}, call that a jet and remove it. One exponent p selects the algorithm — the *same* kernel serves all three.

One exponent, three algorithms

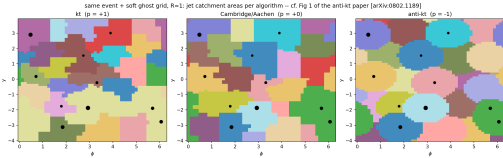
The generalised-k_t distance measure:

$$d_{ij} = \min\left(p_{Ti}^{2p}, p_{Tj}^{2p}\right) \frac{\Delta R_{ij}^2}{R^2}, \quad \Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$$

$$d_{iB} = p_{Ti}^{2p}$$

Merge the pair with the smallest d_{ij}; if the smallest distance is a d_{iB}, call i a jet. **Only p changes:**

p	algorithm	merges first
-1	anti-k_t	around the **hardest** particles
0	Cambridge–Aachen	the **closest** pair
$+1$	k_t	the **softest** pair



anti-k_t catchment areas $\to \pi R^2$ — the defining rigid-cone result, here from flashjet's own clustering.

The ordering is what the substructure reads:

- **C/A** by *angle* $\to$ **grooming**, **Lund**
- k_t by *relative* p_T $\to$ **exclusive subjets**

It returns the tree, not just the jets

Clustering a batch returns **two** things — the jets, and the record of how they were built.

field	shape	what it holds
`jet_idx`	(B, N)	which jet each particle ended up in
`n_jets`	$(B,)$	how many jets were found
`hist_p1`, `hist_p2`	(B, N)	**which pair merged**, at each step
`hist_child`	(B, N)	what they merged *into*
`hist_d`	(B, N)	the distance d_{ij} at that step

Why the second half matters

The bottom three rows are the **complete binary merge tree**. Groomed mass, z_g, R_g, Lund coordinates and exclusive subjets are all **coordinates on that tree** — so they become array reads, with **no second clustering pass**.

Using it

```
import flashjet

out = flashjet.cluster(p4, mask, R=1.0, algorithm="antikt")
```

argument	type	
p4	$(B, N, 4)$ tensor	(p_x, p_y, p_z, E); CPU or CUDA
mask	(B, N) bool	marks real constituents in a padded batch
R	float	jet radius
algorithm	str	"antikt" | "kt" | "cambridge"
p	float	generalised-k_t exponent (overrides algorithm)

```
out.n_jets                                        # (B,)     jets found
out.jets_p4(p4)                                   # (B,J,4)  jet four-momenta
out.splitting_scales()                            #          sqrt(d12), sqrt(d23), ...
out.exclusive_jets(n_jets=3)                      # F1       exclusive kt subjets
out.groomed_jets(p4, R, z_cut=0.1, beta=0.0)      # F2       soft drop / mMDT
out.lund_coordinates(p4, R)                       # F3       primary Lund plane
```

The same call runs unchanged on CPU and GPU — the backend is chosen from the problem size, not by the caller.

What is implemented

Clustering

- generalised-k_t: anti-k_t, k_t, C/A
- batched, ragged input via a mask
- three backends, auto-dispatched by size
- merge history recorded in-kernel
- jet regime *and* event regime

Substructure — reads of the history

- **F1** exclusive k_t subjets
- **F2** soft drop / mMDT / mass-drop
- **F3** primary Lund coordinates

All three are pure-torch post-reads: no kernel changes, CPU/GPU identical, negligible cost.

F1 — exclusive subjets from the k_t history

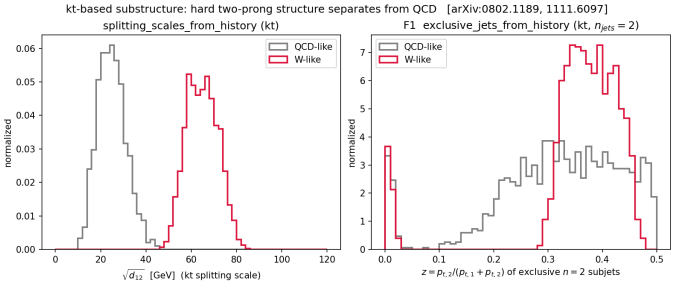
What it does. Undo the last merges of the sequence to expose exactly n subjets — or stop at a d_{cut}.

How. The history is already a binary tree. Walk back up it $n-1$ steps; no reclustering.

Why k_t. k_t merges the softest pair first, so the *last* merges are the hard prongs — exactly what a 2-prong decay leaves behind.

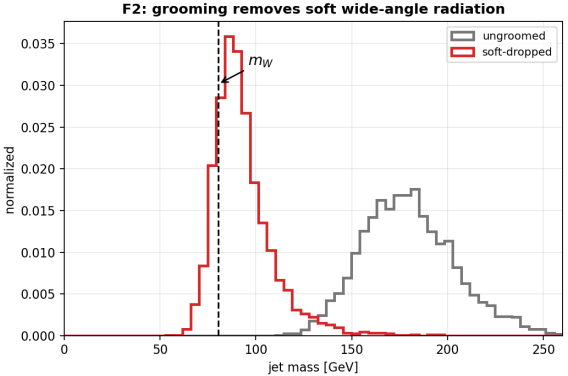
The splitting scales come straight from `hist_d`:

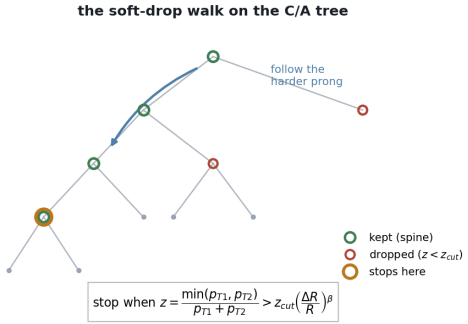
$$\sqrt{d_{12}},\ \sqrt{d_{23}},\ \ldots \quad \text{with} \quad d_{ij} = \min(p_{Ti}^2, p_{Tj}^2)\,\Delta R_{ij}^2/R^2$$



Toy W-like (2-prong) vs QCD-like fat jets. **Left:** the k_t splitting scale $\sqrt{d_{12}}$ separates cleanly. **Right:** the momentum balance z of the two exclusive subjets — the W sits near $z \sim 0.4$, QCD spreads to small z.

F2 — soft drop / mMDT grooming from the C/A history

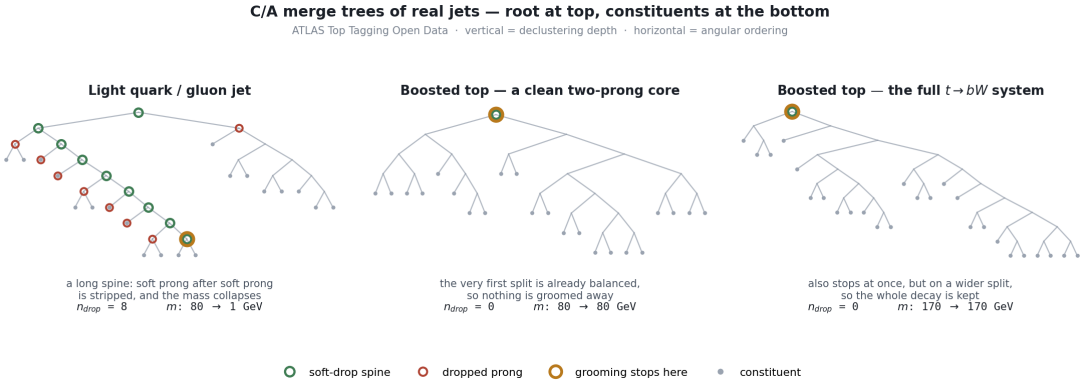




Walk down the C/A tree along the harder branch, discarding the softer prong until the momentum-sharing condition is met. $\beta = 0$ recovers mMDT.

The walk is $O(\log_2 n)$ on the stored tree — it never touches the constituents again, so grooming is essentially free once clustered.

What the tree looks like on real jets

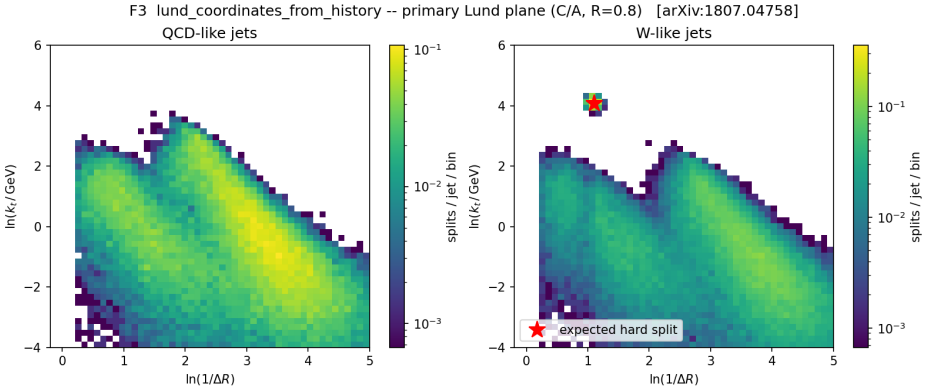


A jet without a decay has nothing to stop on. The light-quark/gluon jet is a long *staircase*: soft prong after soft prong is stripped, and its mass collapses to nothing.

A real decay stops at once. Both top jets halt on their very first split, because it is already hard and balanced — so the mass survives.

ATLAS Open Data, C/A with soft drop ($z_{cut}=0.1$, $\beta=0$). **The number of drops is itself a discriminant** — and it is read straight off the stored tree, at no extra cost.

F3 — primary Lund coordinates from the C/A history



What. Every primary splitting emits $z = \frac{p_{T2}}{p_{T1}+p_{T2}}$, ΔR, $k_t = p_{T2}\Delta R$, and the plane coordinates $\left(\ln\frac{1}{\Delta R},\ \ln k_t\right)$.

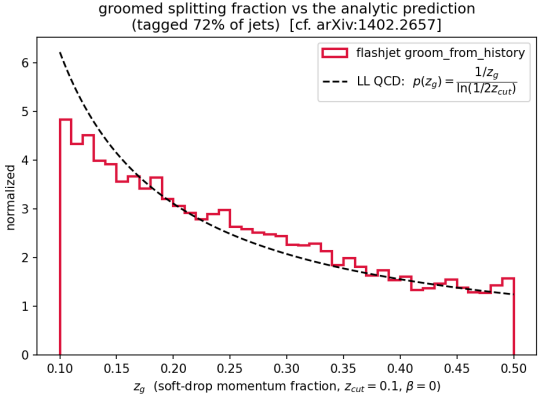
How. One pass down the primary branch — each node is one point in the plane.

Check. The d-channel ties *exactly* to the splitting scales computed independently.

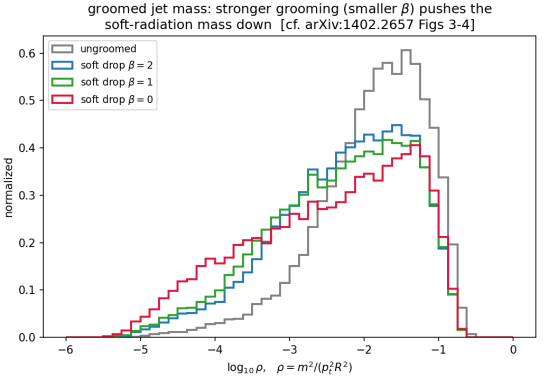
QCD fills the soft-collinear region smoothly; the W-like sample adds a **localised hard splitting** exactly where the decay is predicted to sit (star).

Correctness

The history is right: closures against analytic predictions



Soft-drop z_g follows the $1/z$ form predicted at leading-log, recovered from the C/A history.

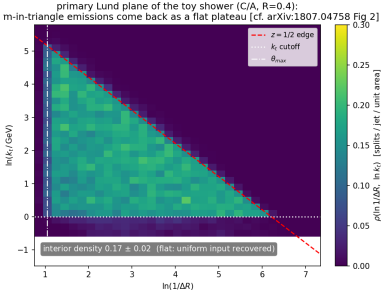


Groomed-mass ordering in β: larger β grooms less, so the mass peak moves up.

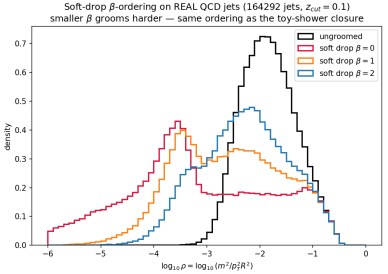
Toy-shower input, so the *prediction* is unambiguous — these test the decoders, not an event generator.

Clustering geometry

anti-k_t jet areas $\to \pi R^2$.



Lund plane uniform inside the kinematic triangle.



β-family ordering.

Independent NumPy tree-walks pin every decoder; the d-channel of the Lund output ties exactly to the splitting scales.

Same jets as FastJet — across the whole grid

3 algorithms $\times$ **5 radii** $\times$ **5 multiplicity bins** $\times$ **2 samples**, 20 000 jets per bin, on ATLAS Open Data.

check	**result**
number of jets found	**100.000 %** at every one of 150 points
leading-jet p_T within 10^{-4}	1.0000 at 148/150 points
median relative difference	$\sim 7 \times 10^{-8}$

k_t and C/A had never been checked against FastJet on real detector data — only against NumPy tree-walks. They now agree across the full radius $\times$ multiplicity grid.

The two exceptions are the *same single jet*: one constituent assigned across an R boundary. Jet counts still match and the leading jet is never ambiguous.

Same jets as the experiment

A stronger test than agreeing with FastJet.

ATLAS publishes an open dataset carrying both the **calorimeter clusters** and **its own reconstructed jets**. Cluster the ∼600 clusters per event ourselves, and compare:

n_{jets} vs ATLAS	**100.0 %** match
mean jets/event	10.96 vs 10.96

This closes the algorithm against **the experiment's own published output** — not against another implementation of the same algorithm.

599.3 clusters/event: the event regime, where the problem is $O(N^2)$.

Speed

The benchmark

Data — ATLAS Open Data, freely presentable

sample	jets	$\langle n_{\text{const}} \rangle$
boosted top	30 000	59.2
QCD (light q/g)	30 000	52.4

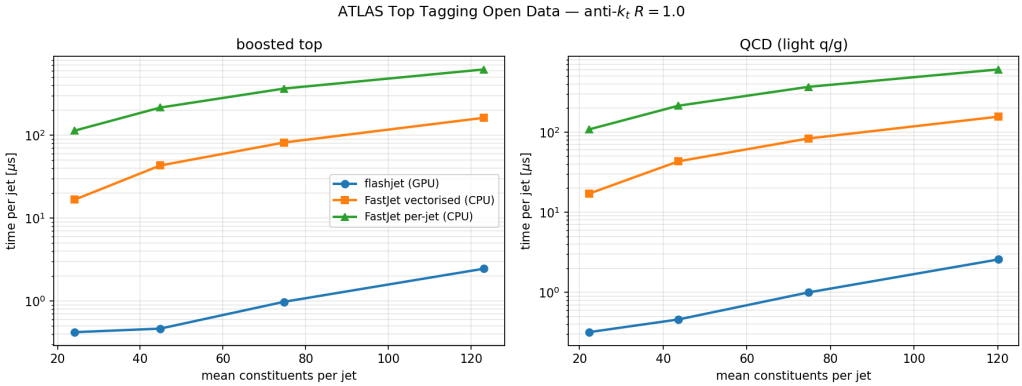
Large-R jets with calorimeter-cluster constituents. The grid is **3 algorithms** $\times$ **5 radii** $\times$ **5 multiplicity bins**, 20 000 jets per bin.

Method

1. extract constituents once, save;
2. cluster on GPU, time it, save the *same* events;
3. cluster those saved events with FastJet;
4. **check they agree** — else the timing is meaningless.

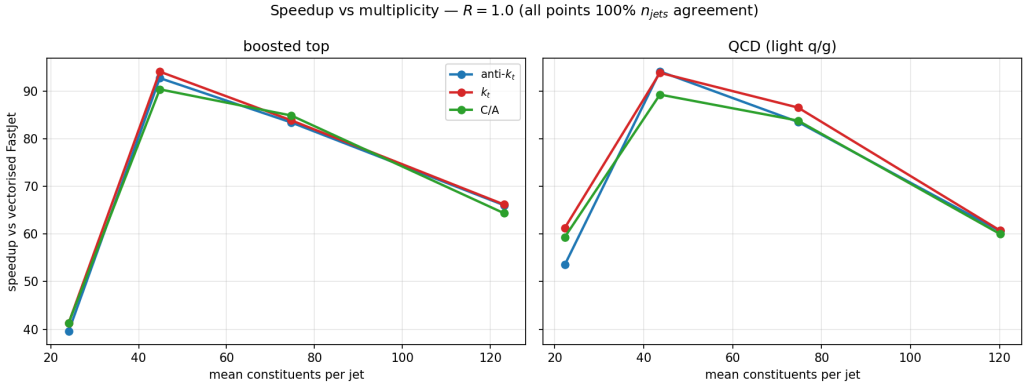
Both clusterers read the **same saved file**. Speedups are quoted against the **vectorised** FastJet interface — the faster, fairer baseline.

Time per jet



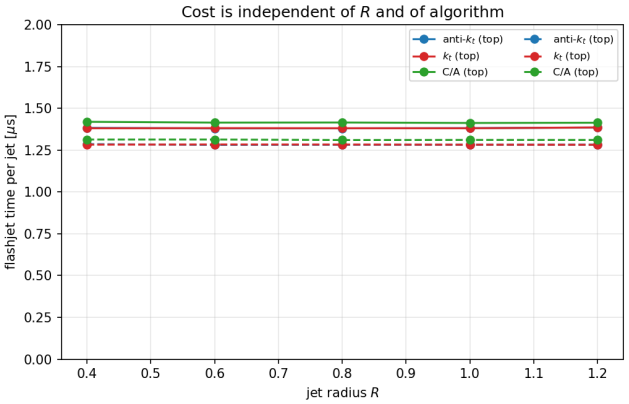
Log scale — each gridline is $10\times$. flashjet stays nearly flat while both FastJet interfaces climb steeply with multiplicity: **0.35–2.7 μs per jet** against **17–155 μs** for vectorised FastJet, and **110–620 μs** per jet one at a time. Benchmarked on an NVIDIA **V100-class** GPU.

The advantage grows with jet complexity



All three algorithms, both samples, $R = 1.0$. **39–99**× over vectorised FastJet across the grid (median **64**×), and every point has **100 %** n_{jets} agreement. Busier jets → bigger win — the regime that matters for boosted-object physics.

Cost is independent of R — and of the algorithm



Across $R = 0.4 \to 1.2$: a **0.3 % spread** over a $3\times$ range in R.

R changes *which* pairs merge, not *how many* distances are computed.

Across algorithms: anti-k_t, k_t and C/A agree to within $\sim$4 % at every multiplicity.

k_t and C/A timing is **free**.

Conclusions

Summary

1. **Clusters on the GPU and keeps the full merge history**, so substructure — exclusive subjets, soft drop, Lund coordinates — comes free as array reads.
2. **Gives the same jets as FastJet.** 150/150 grid points at 100 % n_{jets} agreement, across anti-k_t/k_t/C-A, $R = 0.4$–1.2 and 24–123 constituents per jet. k_t and C/A validated on real data for the first time.
3. **Reproduces the experiment's own reconstructed jets** exactly, at ∼600 clusters per event.
4. **39–99× faster** than vectorised FastJet, and the advantage *grows* with jet complexity. Cost is independent of R and of algorithm.
5. Still a **prototype**, with meaningful headroom left.

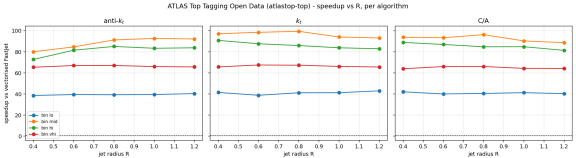
Thank you

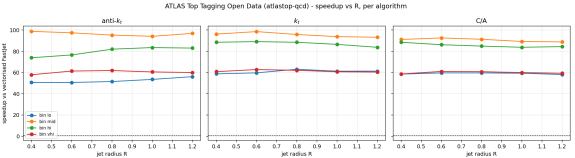
Questions?

Benchmarks: ATLAS Open Data (CC0).
Physics closures: analytic predictions on toy showers.

Backup

Backup — speedup vs R, per algorithm



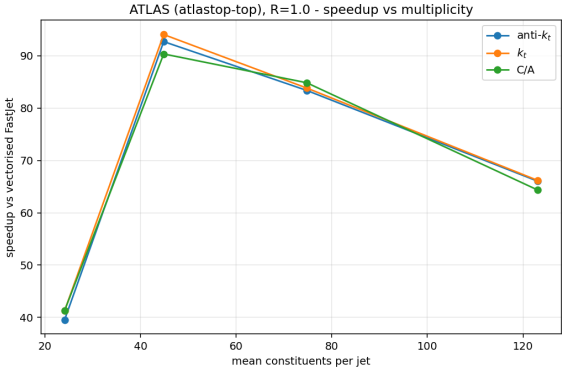


boosted top

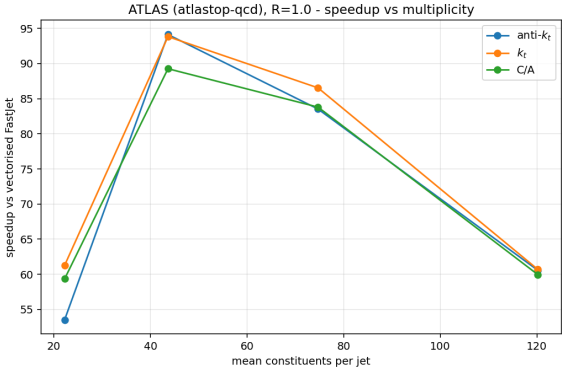
QCD (light q/g)

Flat in R for all three algorithms; ordering is by multiplicity bin.

Backup — speedup vs multiplicity



boosted top



QCD (light q/g)

Backup — the numbers

sample	alg	$\langle n \rangle$	flashjet [μs/jet]	FastJet vec. [μs/jet]	speedup
top	anti-k_t	24.1	0.427	16.5	$39\times$
top	anti-k_t	44.8	0.464	43.0	$\mathbf{93}\times$
top	C/A	74.7	1.000	84.8	$85\times$
top	C/A	123.1	2.565	165.1	$64\times$
qcd	anti-k_t	22.2	0.346	17.5	$51\times$
qcd	anti-k_t	43.6	0.463	45.7	$\mathbf{99}\times$
qcd	anti-k_t	74.7	1.215	89.8	$74\times$
qcd	anti-k_t	120.2	2.673	154.9	$58\times$

39–99$\times$ over vectorised FastJet (median $64\times$), with 100 % jet-count agreement everywhere. Benchmarked on an **NVIDIA V100-class** GPU.

$R = 1.0$ rows shown. Against the per-jet FastJet interface the gap is roughly *two orders of magnitude*.

Backup — the datasets

All benchmark data is **ATLAS Open Data** (CC0), read over the public redirector — no grid certificate required.

dataset	used for
ATLAS Top Tagging Open Data	timing + FastJet agreement
2020 ATLAS Jet Reconstruction	closure vs the experiment's own jets

Constituents are calorimeter clusters, massless. Physics closures use toy showers compared against leading-log analytic predictions, so the prediction being tested is unambiguous.